

Discrete Mathematics - Practical 2-3

Relations: worked solutions
with the reasoning spelled out, step by step

Warm-up: what a relation is, and why we draw it

A (binary) relation ρ on sets is just a *set of ordered pairs*. Writing $(x, y) \in \rho$ (also $x \rho y$) means " x is related to y ". That is the whole definition - but a bare list of pairs hides everything interesting, so a relation is almost always better understood as a **picture**:

- On a finite set, draw a **graph**: a dot for each element and an arrow $x \rightarrow y$ for each pair $(x, y) \in \rho$. If the relation is symmetric we drop the arrowheads and draw plain edges.
- The same data is a 0/1 **adjacency matrix** M : put $M_{ij} = 1$ exactly when (element i , element j) $\in \rho$. The graph and the matrix are two views of one object.
- On $\mathbb{R}^2$ a relation is a **region or curve in the plane**; its domain and range are the shadows it casts on the x - and y -axes.

We use the course's conventions throughout. The **domain** is the set of first coordinates, $D_\rho = \{x : \exists y (x, y) \in \rho\}$, and the **range** is the set of second coordinates, $R_\rho = \{y : \exists x (x, y) \in \rho\}$. **Composition is left-to-right**: $\rho \cdot \sigma$ means "do ρ first, then σ ", i.e. $(x, z) \in \rho \cdot \sigma \iff \exists y : (x, y) \in \rho \text{ and } (y, z) \in \sigma$. (This is the opposite order to the $\circ$ many books use, so watch the direction.) Here $\mathbb{N} = \{1, 2, 3, \dots\}$.

Problem 1.2 - Domain and range

Problem 1.2

Find the domain D_ρ and range R_ρ of each relation.

- | | |
|--|--|
| a) $\{(a, 1), (a, 2), (c, 1), (c, 2), (c, 4), (d, 5)\}$ | e) $\{(x, y) \in \mathbb{N}^2 : y \mid x\}$ (source: x/y) |
| b) $\{(1, 2), (2, 4), (3, 6), (4, 8), \dots\} = \{(n, 2n)\}$ | f) $\{(x, y) \in \mathbb{N}^2 : x \mid y\}$ (source: y/x) |
| c) $\{(x, y) \in \mathbb{R}^2 : x = y^2\}$ | g) $\{(x, y) \in \mathbb{R}^2 : x + y \leq 0\}$ |
| d) $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 16\}$ | h) $\{(x, y) \in \mathbb{R}^2 : 2x \geq 3y\}$ |

Here the sheet's notation x/y is the divisibility predicate " x over y is a whole number", i.e. $y \mid x$ in (e), and $x \mid y$ in (f).

The idea

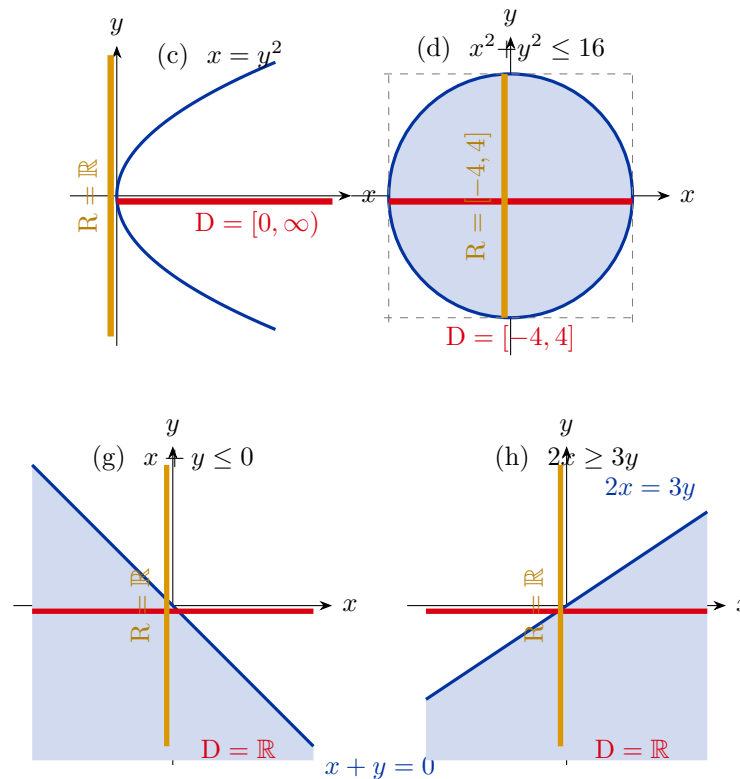
Domain and range are **shadows**. Stand a light to the side and let the relation cast its shadow on the x -axis: that shadow is the domain (which x -values appear as a first coordinate). Shine from below onto the y -axis and the shadow is the range. So you never need the whole relation at once - you only ask, for each candidate x , "**is there at least one y that partners it?**" If yes, x is in the domain. Same question with the roles of x and y swapped gives the range. For the plane relations, draw the region and read the two shadows off the axes; for the discrete ones, a single witnessing partner settles each value.

(a) **Finite list - just read it off.** First coordinates: a, c, d . Second coordinates: $1, 2, 4, 5$.

$$D = \{a, c, d\}, \quad R = \{1, 2, 4, 5\}.$$

(b) **The doubling relation** $\{(n, 2n)\}$. Every $n \in \mathbb{N}$ is a first coordinate, so $D = \mathbb{N}$. The second coordinates are exactly the even numbers, so $R = \{2, 4, 6, \dots\} = 2\mathbb{N}$.

(c), (d), (g), (h) - **the plane relations**. Draw each, then read the shadows.



(c) $x = y^2$. The curve is a parabola opening along the positive x -axis. An x appears iff $x = y^2$ has a solution, i.e. iff $x \geq 0$; so $D = [0, \infty)$. Any real y partners the point $x = y^2$, so every y appears: $R = \mathbb{R}$.

(d) **the disk** $x^2 + y^2 \leq 16$. Radius $\sqrt{16} = 4$. A value x appears iff some y has $x^2 + y^2 \leq 16$; the easiest partner is $y = 0$, which works iff $x^2 \leq 16$, i.e. $x \in [-4, 4]$. By symmetry $R = [-4, 4]$ too.

(g) $x + y \leq 0$. The region is the closed half-plane on and below the line $x + y = 0$. Given any x , pick $y \leq -x$ (e.g. $y = -x$); given any y , pick $x \leq -y$. Every real works on both sides, so $D = R = \mathbb{R}$.

(h) $2x \geq 3y$. Half-plane on and below the line $2x = 3y$ (that is $y = \frac{2}{3}x$). Given any x , take $y \leq \frac{2x}{3}$; given any y , take $x \geq \frac{3y}{2}$. Again $D = R = \mathbb{R}$.

(e), (f) - **divisibility on** $\mathbb{N}$. Recall $y \mid x$ means y divides x . For (e) ($y \mid x$): every $x \in \mathbb{N}$ is divisible by $y = 1$, so x always has a partner, $D = \mathbb{N}$; and every y divides $x = y$ itself, so y always appears, $R = \mathbb{N}$. For (f) ($x \mid y$) the same argument with x, y swapped gives $D = \mathbb{N}$ and $R = \mathbb{N}$.

Always check. The Python enumeration over $1 \leq x, y \leq 199$ for (e),(f) produced every value $1, \dots, 199$ in both the domain and the range, and the bounded plane samples confirmed $D_c = [0, \infty)$, $D_d = R_d = [-4, 4]$, and $D = R = \mathbb{R}$ for (g),(h). ✓

	D	R
a)	$\{a, c, d\}$	$\{1, 2, 4, 5\}$
b)	$\mathbb{N}$	$2\mathbb{N} = \{2, 4, 6, \dots\}$
c)	$[0, \infty)$	$\mathbb{R}$
d)	$[-4, 4]$	$[-4, 4]$
e)	$\mathbb{N}$	$\mathbb{N}$
f)	$\mathbb{N}$	$\mathbb{N}$
g)	$\mathbb{R}$	$\mathbb{R}$
h)	$\mathbb{R}$	$\mathbb{R}$

Answer: See the table above: domains and ranges for (a) to (h).

★ Method to remember

Transferable technique: domain and range are the two axis-shadows of the relation. To test if a value x_0 is in the domain you do not need the whole picture - just exhibit *one* partner y with $(x_0, y) \in \rho$. For relations in $\mathbb{R}^2$, sketch the region and literally read the shadows; for an equation $x = f(y)$ the domain is the set of outputs of f and the range is f 's input set.

Problem 2.1 - Undirected graph and adjacency matrix

Problem 2.1

On $A = \{a, b, c, d, e\}$ build the graph and the adjacency matrix of each relation (rows and columns ordered a, b, c, d, e).

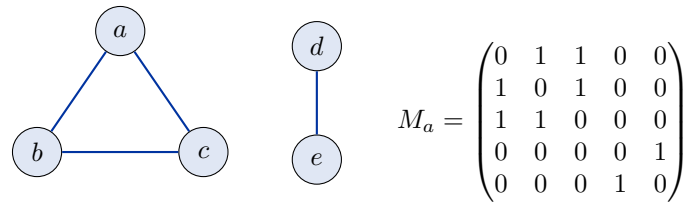
- a) $\{(a, b), (b, a), (b, c), (c, b), (c, a), (a, c), (d, e), (e, d)\}$
- b) $\{(a, b), (b, a), (b, c), (c, b), (c, d), (d, c), (c, a), (a, c)\}$
- c) same set as (b) in the source.

The idea

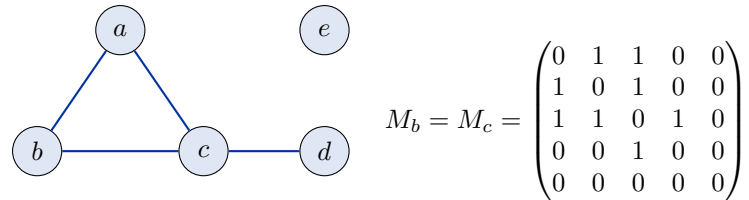
Each pair (i, j) in the relation is one 1 in the matrix (row i , column j) and one arrow $i \rightarrow j$ in the graph. Here every relation is **symmetric**: whenever (i, j) is present so is (j, i) . Symmetry shows up two ways at once - in the matrix as $M = M^T$ (it is mirror-image across the diagonal), and in the graph as "every arrow has its reverse", so we may erase the arrowheads and draw a single undirected **edge** per mutual pair. Reading the picture is then just counting edges; reading the matrix is just checking the mirror.

Step 1 - List the undirected edges. (a) The mutual pairs give edges $a-b$, $b-c$, $a-c$ (a triangle on a, b, c) and a separate edge $d-e$. (b) Edges $a-b$, $b-c$, $a-c$ (triangle again) plus $c-d$; vertex e has no edge. (c) is identical to (b).

Step 2 - Draw and write the matrix side by side.



(a) triangle abc + edge de



(b) = (c) triangle abc + edge cd ; e isolated

Always check. Each matrix is mirror-symmetric across the main diagonal (as it must be for a symmetric relation), the diagonal is all 0 (no pair (x, x) is listed), and the number of 1s equals twice the number of edges: (a) 8 ones = 2×4 edges, (b) 8 ones = 2×4 edges. The Python check confirmed both matrices entry-for-entry and that each is symmetric. ✓

Answer: (a) triangle abc plus edge de ; (b)=(c) triangle abc plus edge cd , e isolated.
Matrices as drawn.

★ Method to remember

Transferable technique: a symmetric relation $\Leftrightarrow$ an undirected graph $\Leftrightarrow$ a symmetric 0/1 matrix. Count of 1s = $2 \cdot (\text{\#edges})$, and the diagonal records the loops (x, x) . To sanity-check a hand-drawn graph against its matrix, verify $M = M^T$.

Problem 2.2 - Directed graph (digraph) and adjacency matrix

Problem 2.2

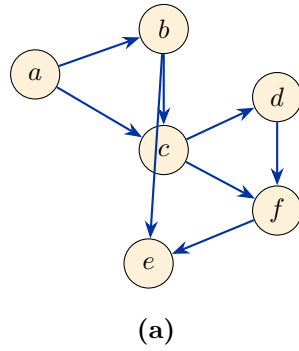
On $A = \{a, b, c, d, e, f\}$ build the digraph and the adjacency matrix of each relation (order a, b, c, d, e, f). An arrow goes $i \rightarrow j$ exactly when $(i, j) \in \rho$.

- $\{(a, b), (b, c), (a, c), (b, e), (c, f), (c, d), (d, f), (f, e)\}$
- $\{(a, b), (b, c), (d, c), (d, e), (f, e), (f, a), (b, e)\}$
- $\{(a, c), (c, c), (c, d), (c, e), (e, c), (e, e), (e, f), (a, e)\}$

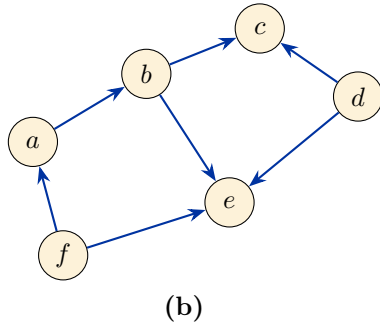
The idea

A general relation need not be symmetric, so we keep the **arrowheads**: (i, j) is an arrow from i to j , and it need not come with the reverse arrow $j \rightarrow i$. In the matrix this means M is no longer mirror-symmetric: row i tells you where you can go from i , column j tells you what points to j . A pair (x, x) is a **self-loop** at x and a 1 on the diagonal. Drawing the digraph is the fastest way to see structure the list hides ("is there a path from a to f ? a cycle?").

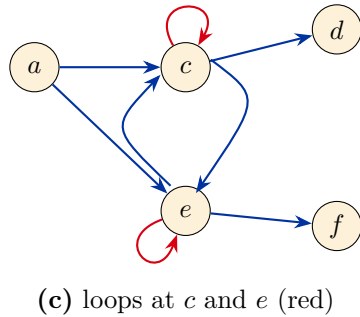
Read the matrices off the lists (row = source, column = target):



$$M_a = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$



$$M_b = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$



$$M_c = \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Always check. Count the arrows in each list: (a) 8, (b) 7, (c) 8; these equal the number of 1s in M_a, M_b, M_c respectively. The two diagonal 1s of M_c are exactly the loops (c, c) and (e, e) we drew in red. The Python check confirmed all three matrices entry-for-entry and the loop positions. ✓

Answer: Digraphs as drawn; matrices M_a, M_b, M_c as shown. Only (c) has self-loops, at c and e.

★ Method to remember

Transferable technique: for a general (directed) relation, $M_{ij} = 1 \iff i \rightarrow j$. Asymmetry of the relation = asymmetry of M ; a self-loop = a 1 on the diagonal. *Row i* lists the out-neighbours of i ; *column j* lists the in-neighbours of j . A fast cross-check: number of arrows = number of 1s.

Problem 3.3 - Composition of two relations

Problem 3.3

On $\mathbb{Z}$ let $\rho = \{(x, x+2) : x \in \mathbb{Z}\}$ (the map $x \mapsto x+2$) and $\sigma = \{(x, x^2) : x \in \mathbb{Z}\}$ (the map $x \mapsto x^2$). Find $\rho \cdot \sigma$ and $\sigma \cdot \rho$.

The idea

Composition chains two relations through a **shared middle value**. With the course's left-to-right rule, $\rho \cdot \sigma$ means "do ρ , then σ ": start at x , let ρ carry you to some y , then let σ carry y onward to z . The middle y is the hinge; the output relation pairs the original x with the final z . Because both ρ and σ are *functions* here (each input has exactly one output), the composition is just **function composition** - and the order matters, because adding 2 then squaring is not the same as squaring then adding 2.

Step 1 - $\rho \cdot \sigma$ (apply ρ first). Start at x . ρ sends $x \mapsto y = x+2$. Then σ sends $y \mapsto z = y^2 = (x+2)^2$. So

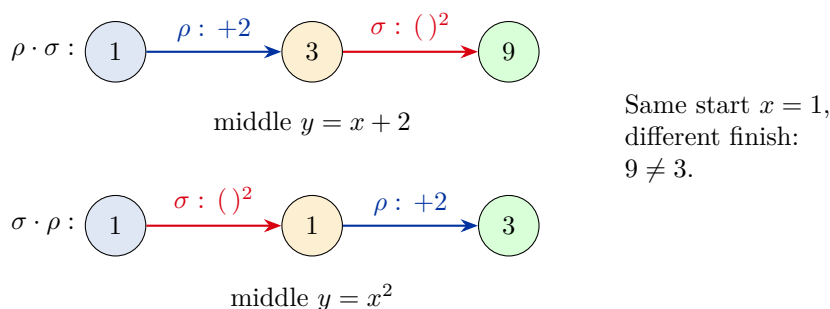
$$\rho \cdot \sigma = \{(x, (x+2)^2) : x \in \mathbb{Z}\}.$$

Step 2 - $\sigma \cdot \rho$ (apply σ first). Start at x . σ sends $x \mapsto y = x^2$. Then ρ sends $y \mapsto z = y+2 = x^2+2$. So

$$\sigma \cdot \rho = \{(x, x^2+2) : x \in \mathbb{Z}\}.$$

Step 3 - they are different. $(x+2)^2 = x^2 + 4x + 4 \neq x^2 + 2$ in general. A single value proves it: at $x = 1$, $\rho \cdot \sigma$ gives $(1, 9)$ while $\sigma \cdot \rho$ gives $(1, 3)$.

The arrow diagram. Tracking $x = 1$ through the shared middle value makes the asymmetry visible:



Always check. The Python enumeration over $-20 \leq x \leq 20$ gave $\rho \cdot \sigma$ output $(x+2)^2$ and $\sigma \cdot \rho$ output x^2+2 for every x , agreeing with the formulas, and found them unequal (first disagreement already at $x = 1$). ✓

Answer: $\rho \cdot \sigma = \{(x, (x+2)^2) : x \in \mathbb{Z}\}$ and $\sigma \cdot \rho = \{(x, x^2+2) : x \in \mathbb{Z}\}$; they are *not* equal (e.g. $(1, 9)$ vs $(1, 3)$).

★ Method to remember

Transferable technique: to compose left-to-right, push a generic element through the *first* relation, then the *second*, naming the middle value. For function-relations $\rho \cdot \sigma$ is the function " σ after ρ ", so $(\rho \cdot \sigma)(x) = \sigma(\rho(x))$. Composition is generally **not commutative** - always test with one concrete input.

Problem 4.1 - Basic properties of relations

Problem 4.1

Decide which of reflexive (R), symmetric (S), antisymmetric (AS), transitive (T) each relation has.

- a) $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$
- b) $\{(x, y) \in \mathbb{R}^2 : x^2 = y^2\}$
- c) $\{(x, y) \in \mathbb{Z}^2 : x/y\}$ (divisibility: $y \mid x$, i.e. " x over y is an integer")
- d) $\{(x, y) \in \mathbb{N}^2 : x/y\}$ (same divisibility, on $\mathbb{N}$)
- e) $\{(x, y) \in \mathbb{N}^2 : (x, y) = 1\}$ (coprime: $\gcd(x, y) = 1$)
- f) $\{(x, y) \in \mathbb{Z}^2 : (x, y) = 1\}$ (coprime, on $\mathbb{Z}$)

Caution: the source writes the divisibility predicate as x/y in (c),(d) - this is *not* the order relation $x \leq y$. Below we treat (c),(d) as divisibility, as the sheet states.

The idea

The four properties are four small **shapes you look for in the digraph**. **Reflexive**: a loop at every vertex (the diagonal is full). **Symmetric**: every arrow comes with its reverse (so it could be drawn undirected). **Antisymmetric**: *no* 2-cycle between distinct vertices - if $x \rightarrow y$ and $y \rightarrow x$ then $x = y$ (the one exception allowed is a loop). **Transitive**: every 2-step path is short-circuited by a direct arrow ($x \rightarrow y \rightarrow z$ forces $x \rightarrow z$). To *disprove* a property you only need one bad picture; to *prove* it you argue in general. The named combinations are worth memorising: $R+S+T$ = **equivalence**; $R+AS+T$ = **partial order**.

(a) $x^2 + y^2 = 1$ on $\mathbb{R}^2$ (the unit circle).

- *Not reflexive*: (x, x) needs $2x^2 = 1$, true only at $x = \pm \frac{1}{\sqrt{2}}$, not for all x .
- *Symmetric*: $x^2 + y^2$ is unchanged when x, y swap, so $(x, y) \in \rho \Rightarrow (y, x) \in \rho$.
- *Not antisymmetric*: $(1, 0)$ and $(0, 1)$ are both in ρ but $1 \neq 0$.
- *Not transitive*: $(1, 0), (0, 1) \in \rho$ but $(1, 1)$ needs $1 + 1 = 1$, false.

$\Rightarrow$ symmetric only.

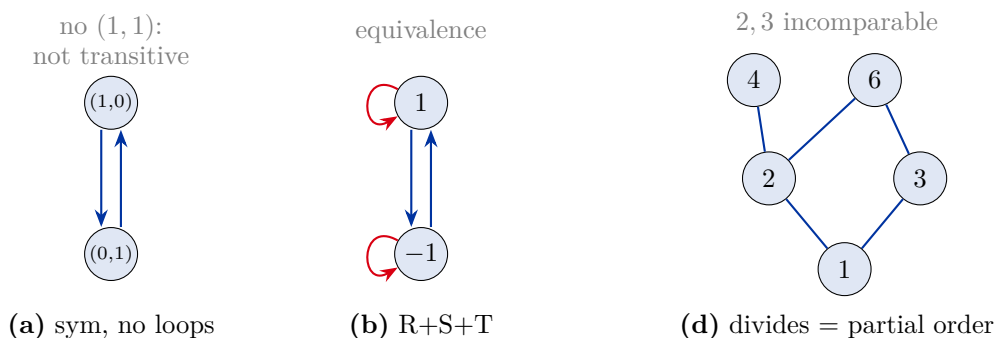
(b) $x^2 = y^2$ on $\mathbb{R}^2$, i.e. $|x| = |y|$. Reflexive ($x^2 = x^2$); symmetric ($x^2 = y^2 \iff y^2 = x^2$); transitive ($x^2 = y^2 = z^2$). Not antisymmetric: $(1, -1)$ and $(-1, 1)$ are both in ρ with $1 \neq -1$. $R+S+T \Rightarrow$ **equivalence relation** (its classes are the pairs $\{x, -x\}$).

(c) **divisibility on $\mathbb{Z}$ ($y \mid x$)**. Reflexive ($x \mid x$); transitive ($z \mid y, y \mid x \Rightarrow z \mid x$). Not symmetric ($2 \mid 4$ but $4 \nmid 2$). *Not* antisymmetric on $\mathbb{Z}$: $2 \mid (-2)$ and $(-2) \mid 2$, yet $2 \neq -2$. $R+T$ but not AS, not $S \Rightarrow$ a **preorder**, *not* a partial order.

(d) **divisibility on $\mathbb{N}$ ($y \mid x$)**. Reflexive and transitive as above; not symmetric. Now it *is* antisymmetric: on positive integers, $a \mid b$ and $b \mid a$ force $a = b$. $R+AS+T \Rightarrow$ **partial order**. It is not a *total* order: 2 and 3 are incomparable (neither divides the other).

(e),(f) **coprimality $\gcd(x, y) = 1$** . Symmetric ($\gcd$ is symmetric). Not reflexive: $\gcd(2, 2) = 2 \neq 1$. Not antisymmetric: $(2, 3)$ and $(3, 2)$ both coprime, $2 \neq 3$. Not transitive: $\gcd(2, 3) = 1$ and $\gcd(3, 4) = 1$ but $\gcd(2, 4) = 2 \neq 1$. So symmetric only, on both $\mathbb{N}$ (e) and $\mathbb{Z}$ (f).

Pictures that make the verdicts visible. (a) a 2-cycle with no loops and a broken triangle; (b) loops everywhere with mirrored arrows; (d) divisibility on $\{1, 2, 3, 4, 6\}$ drawn as a Hasse-style order; (e) a 2-cycle, again no loops.



Always check. A brute-force scan computed all four properties directly from their definitions on finite samples ($\mathbb{R}$ cases on points including the relevant circle/sign points; (c) on $\{-6, \dots, 6\} \setminus \{0\}$; (d) on $\{1, \dots, 12\}$; (e),(f) up to 14). Every verdict below matched, including AS failing for (c) on $\mathbb{Z}$ via 2 and -2 , and AS holding for (d) on $\mathbb{N}$. ✓

Relation	R	S	AS	T	Conclusion
a) $x^2 + y^2 = 1$ on $\mathbb{R}^2$	×	✓	×	×	symmetric only
b) $x^2 = y^2$ on $\mathbb{R}^2$	✓	✓	×	✓	equivalence
c) divisibility on $\mathbb{Z}^2$	✓	×	×	✓	preorder (not antisym.)
d) divisibility on $\mathbb{N}^2$	✓	×	✓	✓	partial order (not total)
e) $\gcd(x, y) = 1$ on $\mathbb{N}^2$	×	✓	×	×	symmetric only
f) $\gcd(x, y) = 1$ on $\mathbb{Z}^2$	×	✓	×	×	symmetric only

Answer: a) S only; b) equivalence (R,S,T); c) divisibility on $\mathbb{Z}$: R,T but not AS (preorder); d) divisibility on $\mathbb{N}$: partial order (R,AS,T, not total); e),f) S only.

★ Method to remember

Transferable technique: test each property as a picture - reflexive = all loops, symmetric = all arrows reversible, antisymmetric = no distinct 2-cycle, transitive = shortcuts present. *One* counterexample kills a property. Two famous packages: R+S+T = equivalence, R+AS+T = partial order. The same predicate can change class with the ground set: divisibility is a partial order on $\mathbb{N}$ but only a preorder on $\mathbb{Z}$ (because n and $-n$ divide each other).

Problem 7.2 - Reflexive, symmetric, transitive closures

Problem 7.2

On $A = \{1, 2, 3, 4\}$ let $\rho = \{(1, 2), (3, 1), (3, 2), (2, 4)\}$. Find the reflexive closure $r(\rho)$, the symmetric closure $s(\rho)$, and the transitive closure $t(\rho)$.

The idea

A **closure** is the *smallest* enlargement of ρ that gains a desired property while adding nothing unnecessary. Each closure has a one-line recipe that says exactly which arrows are

"forced":

$$r(\rho) = \rho \cup \Delta_A \quad (\text{add every loop}), \quad s(\rho) = \rho \cup \rho^{-1} \quad (\text{add every reverse}),$$

$$t(\rho) = \bigcup_{k \geq 1} \rho^k \quad (\text{add an arrow for every directed path}).$$

Reading them on the digraph: r stamps a loop on each vertex; s makes every arrow two-way; t asks "where can I get to by following arrows?" and draws a direct arrow to every reachable vertex. Closures only *add* edges - the original arrows always stay.

Step 1 - Draw ρ and find what each closure must add. Reachability in ρ : $1 \rightarrow 2 \rightarrow 4$, and $3 \rightarrow 1 \rightarrow 2 \rightarrow 4$ (also $3 \rightarrow 2 \rightarrow 4$). So the new *transitive* arrows are $1 \rightarrow 4$ (path $1 \rightarrow 2 \rightarrow 4$) and $3 \rightarrow 4$ (path $3 \rightarrow 1 \rightarrow 2 \rightarrow 4$); note $3 \rightarrow 2$ is already present.

Step 2 - Apply the three recipes.

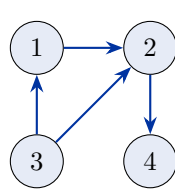
$$r(\rho) = \rho \cup \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

$$= \{(1, 2), (3, 1), (3, 2), (2, 4), (1, 1), (2, 2), (3, 3), (4, 4)\}.$$

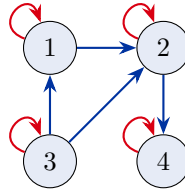
$$s(\rho) = \rho \cup \rho^{-1} = \{(1, 2), (2, 1), (3, 1), (1, 3), (3, 2), (2, 3), (2, 4), (4, 2)\}.$$

$$t(\rho) = \rho \cup \{(1, 4), (3, 4)\} = \{(1, 2), (2, 4), (3, 1), (3, 2), (1, 4), (3, 4)\}.$$

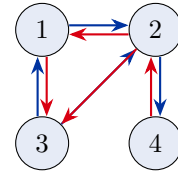
Before / after on the digraph (original arrows in blue, **added arrows in red**):



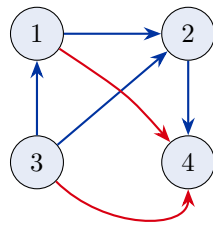
ρ (given)



$r(\rho)$: +loops



$s(\rho)$: +reverses



$t(\rho)$: + (1, 4) via $1 \rightarrow 2 \rightarrow 4$, + (3, 4) via $3 \rightarrow \dots \rightarrow 4$

Reachability:

$1 \rightarrow 2 \rightarrow 4$

$3 \rightarrow 1 \rightarrow 2 \rightarrow 4$

so add (1, 4) and (3, 4).

Always check. The transitive closure was recomputed by Warshall's algorithm on the four nodes; it returned exactly $\{(1, 2), (1, 4), (2, 4), (3, 1), (3, 2), (3, 4)\}$, matching the by-hand answer. The reflexive and symmetric closures were verified by directly applying $\rho \cup \Delta_A$ and $\rho \cup \rho^{-1}$. ✓

$$\begin{aligned} \text{Answer: } r(\rho) &= \rho \cup \{(1, 1), (2, 2), (3, 3), (4, 4)\}; \\ s(\rho) &= \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2), (2, 4), (4, 2)\}; \\ t(\rho) &= \{(1, 2), (2, 4), (3, 1), (3, 2), (1, 4), (3, 4)\}. \end{aligned}$$

★ Method to remember

Transferable technique: closures have fixed recipes - $r = \rho \cup \Delta$, $s = \rho \cup \rho^{-1}$, $t = \bigcup_k \rho^k$ - and each only *adds* edges. For t , the clean way is reachability: draw an arrow $x \rightarrow z$ whenever a directed path goes from x to z (Warshall's algorithm mechanises exactly this). On a digraph: r stamps every loop, s doubles every arrow, t shortcuts every path.